FRONTEND LAYER

- Vanilla JavaScript
- HTML5
- CSS3
- Server-Sent Events (SSE) (real-time updates)

BACKEND LAYER

- Python 3.10+
- Flask (API + server)
- Threading / Async Workers (concurrency model)

AI / ML STACK

Vision & Similarity

- OpenAI CLIP (perceptual embeddings)
- PyTorch (CLIP backend)

Object Detection

- YOLOv8 (v8n)
- Ultralytics YOLO framework

LLM / Reasoning

- Google Gemini 1.5 Flash
- Structured Prompting Layer
- Fallback Heuristic Engine (Local)

VECTOR & DATA LAYER

Vector Database

- Qdrant
- HNSW Indexing
- Cosine Similarity Search

Metadata Database

- MongoDB

DISCOVERY & CRAWLING STACK

- Google Search API
- Bing Search API
- Social Media APIs (platform-dependent)
- yt-dlp (metadata extraction)
- Parallel HTTP Requests (multi-source crawling)

SECURITY STACK

Application Security

- Anti-Debugging Detection
- Environment Gating (.env validation)
- Filename Sanitization (Regex)
- Rate Limiting

Data Security

- Vector Anonymization (no raw media)
- Access Control via Environment Flags

INFRASTRUCTURE & DEVOPS

Containerization

- Docker
- Docker Compose

WRAITH ASSET PROTECTION SYSTEM – WORKFLOW

1. MAIN SCAN WORKFLOW (THE "SCAN" JOURNEY)

1. User Input
2. URL Submit
3. Flask Validate
4. Async Worker
5. Frame Extract
6. CLIP Vectors
7. Vector Search
8. Qdrant Match
9. Threshold Check
   - No → No Match End
   - Yes →
10. Top Frames
11. YOLO Detect
12. Logo Confirm
13. Gemini Reason
14. Report Generate
15. DMCA Create
16. SSE Stream
17. UI Display

2. GLOBAL DISCOVERY PIPELINE

1. Topic Input
2. Query Expand
3. Multi Crawl
4. Risk Score
5. High Risk → Auto Queue

3. DATA LAYER

MongoDB (dap_db)

- Clips
- Jobs
- Reports
- Discovery

Qdrant (video_frames)

- Vector Data
- HNSW Index

4. SECURITY LAYER

- Debug Detect
- Env Gate
- File Sanitize
- Rate Limit
- Vector Protect

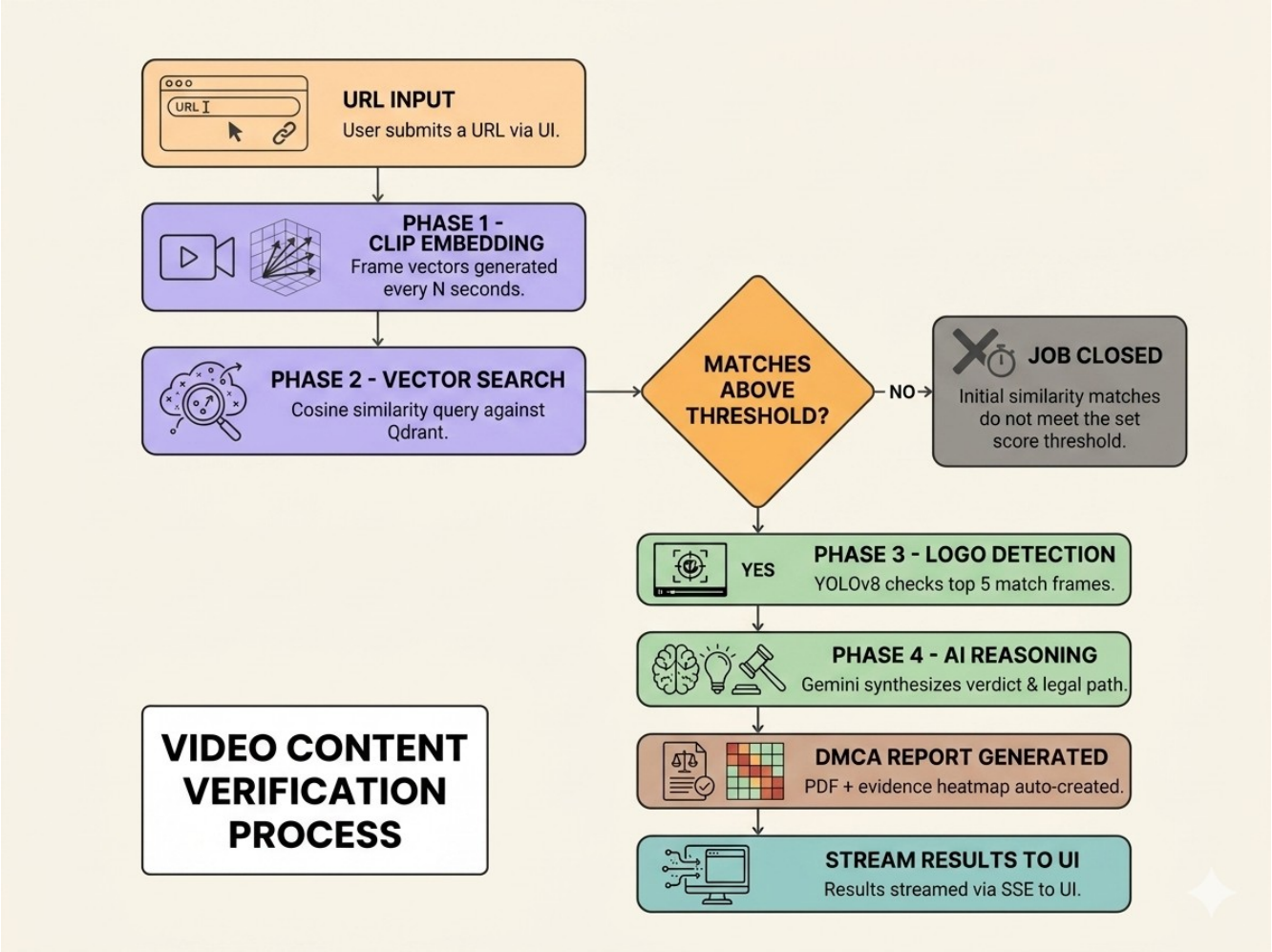
5. INFRASTRUCTURE & DEPLOYMENT

Docker Deploy → Service Start

- Flask Server
- Worker Pool
- Qdrant DB
- Mongo DB

6. OUTPUT & DELIVERY

Real-time Updates → Scan Results → DMCA Report → Enforcement Ready


URL INPUT
User submits a URL via UI.
PHASE 1 - CLIP EMBEDDING
Frame vectors generated every N seconds.
PHASE 2 - VECTOR SEARCH
Cosine similarity query against Qdrant.
MATCHES ABOVE THRESHOLD?
NO
JOB CLOSED
Initial similarity matches do not meet the set score threshold.
YES
PHASE 3 - LOGO DETECTION
YOLOv8 checks top 5 match frames.
PHASE 4 - AI REASONING
Gemini synthesizes verdict & legal path.
DMCA REPORT GENERATED
PDF + evidence heatmap auto-created.
STREAM RESULTS TO UI
Results streamed via SSE to UI.
VIDEO CONTENT VERIFICATION PROCESS